

topcatswap

Installation and hosting guide

Web app, bundle relay, Ubuntu desktop app and command-line bot

Ubuntu 22.04 / 24.04 · Node.js 18+ · GitHub Pages / Cloudflare

Web 1.1.3 · Desktop 1.1.2 · English (US)

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1 Options

Option	What you need
Use the web app online	Nothing: open https://topcatmasterinvestor.github.io/topcatswap/
Run the web app on your computer	Python 3 (comes with Ubuntu) or Node.js 18+
Host your own copy	A GitHub or Cloudflare account (free)
Bundle relay for the web	A Cloudflare account (free), or Node.js on your computer
Ubuntu desktop app	Ubuntu 22.04/24.04, Node.js 18+, sudo
Command-line bot	Node.js 18+

Get the code once (needs git: sudo apt install -y git):

```
mkdir -p ~/projetos && cd ~/projetos
git clone https://github.com/TOPCATMASTERINVESTOR/topcatswap.git
cd topcatswap
```

2 Run the web app on your computer

```
cd ~/projetos/topcatswap
node tools/build-web.js
python3 -m http.server 8080 --bind 127.0.0.1 -d dist-web
```

Open http://localhost:8080 in Chrome, Edge or Brave. Always use the localhost address: the Ledger and the extensions need http://localhost or https://. Stop it with Ctrl+C.

> build-web.js needs no npm install: it bundles the same engine as the desktop app with the browser adapters from web-src/.

3 Host your own copy

GitHub Pages

1. Create a repository on GitHub and push the code (git remote add origin ..., git push -u origin main).
2. In the repository: Settings › Pages › Source: GitHub Actions.
3. Every push to main rebuilds and publishes the site at https://<user>.github.io/<repo>/.

Cloudflare Pages

Build with node tools/build-web.js and upload the dist-web folder in Workers & Pages › Create › Pages › Upload assets.

Before publishing

Edit web-src/config.js and rebuild: defaultRelay (your relay address, for all users) and trezorManifest.email (a contact email, required by Trezor Connect).

> On <user>.github.io all your Pages sites share one browser storage. If people will keep imported wallets on your site, use your own domain or a dedicated GitHub account.

Applying updates

```
cd ~/projetos/topcatswap
git pull            # or: git am ~/Downloads/<update>.patch
git push
```

4 Bundle relay (web)

Builders reject requests from websites (CORS). The relay forwards already-signed bundles to a closed list of builders (relay/allowlist.mjs). It stores nothing and never sees keys.

Cloudflare Workers (free)

```
cd ~/projetos/topcatswap/relay
npx wrangler login
npx wrangler deploy
```

Copy the address it prints (https://topcatswap-relay.<account>.workers.dev) into Settings › Connections › Bundle relay, or into web-src/config.js. Recommended: in relay/wrangler.toml set ALLOWED_ORIGINS to your site so only it can use the relay.

On your computer

```
cd ~/projetos/topcatswap
node relay/server.mjs      # http://127.0.0.1:8787
```

5 Ubuntu desktop app

```
cd ~/projetos/topcatswap/desktop
./build-ubuntu.sh
ls dist/
sudo apt install ./dist/TopCatSwap-1.1.2-amd64.deb
topcatswap
```

build-ubuntu.sh installs any missing system packages (asks for the sudo password), the dependencies (Electron, ethers, Ledger and Trezor libraries) and creates the .deb and the AppImage in dist/. Use the exact file name shown by ls dist/.

- AppImage instead of .deb: chmod +x dist/TopCatSwap-*.AppImage && ./dist/TopCatSwap-*.AppImage --no-sandbox (may need sudo apt install libfuse2, or libfuse2t64 on 24.04).
- Update: git pull, then build and install again. Settings and imported wallets in ~/.config/TopCatSwap are kept.
- Uninstall: sudo apt remove topcatswap.

6 Command-line bot

```
cd ~/projetos/topcatswap/desktop
npm install ethers @ledgerhq/hw-transport-node-hid @ledgerhq/hw-app-eth
node engine/topcatswap.en.js --version
node engine/topcatswap.en.js POOL ETH USDC
node engine/topcatswap.en.js SWAP ETH USDC 0.01 --protect
```

Without arguments it shows the full help. engine/topcatswap.js is the Portuguese version. Optional environment variables: RPC_URL, RPC_URL_42161, RPC_URL_56, GRAPH_API_KEY.

7 Hardware wallet preparation

Ledger

```
wget -q -O - https://raw.githubusercontent.com/LedgerHQ/udev-rules/master/add_udev_rules.sh |
sudo bash
```

On the Ledger: Ethereum app › Settings › turn on “Blind signing”. Close Ledger Live when using TopCatSwap. For Bluetooth on Android: Nano X, Stax or Flex with Bluetooth on.

Trezor

```
sudo curl -fsSL https://data.trezor.io/udev/51-trezor.rules -o /etc/udev/rules.d/51-trezor.rul
es
sudo udevadm control --reload-rules && sudo udevadm trigger
```

Close Trezor Suite when using TopCatSwap.

8 Verifying what you run

- Website: each build writes dist-web/SHA256SUMS.txt. Rebuild from the code and compare with the published site's SHA256SUMS.txt.
- Releases: GitHub Releases (created from a v* tag) include the .deb, the AppImage, the website zip and SHA256SUMS.txt. Check with sha256sum -c SHA256SUMS.txt.
- Tests that run on every push: relay, Ledger USB and Bluetooth protocol (simulated device) and the bundle flow.

9 Installation problems

Symptom	Fix
git: command not found	sudo apt install -y git
git push asks for a password	Use gh auth login --web (sudo apt install gh), or a GitHub token with repo and workflow scopes.
Site shows 404 on GitHub Pages	Settings › Pages › Source: GitHub Actions, then wait for the workflow in the Actions tab.
Unsupported file ...deb	The .deb has another name or the build failed: run ls dist/ and use the name shown.
Build compile error	sudo apt install build-essential libudev-dev libusb-1.0-0-dev python3, then build again.
Ledger/Trezor not detected	Install the udev rules, close Ledger Live/Trezor Suite, unplug and plug the cable.
Bundles: relay error	Check the relay address in Settings and, if set, ALLOWED_ORIGINS in wrangler.toml.